

Automation and Utilities



Overview

- Schedule.
- Task Manager.
- %ZSTART and %ZSTOP.
- \$SYSTEM.
- ^OPER.



Schedule

- Daily

- Review size, %Full and status for databases.
- Check disk has enough space for databases and journal files to grow.
- Review system statistics throughout day.
- Review logs.
- Make sure all system daemons are running.
- Verify backups and journaling are running properly.
- Look for process memory images (Troubleshooting module).

- Weekly

- Check integrity of databases and/or specific globals.



Task Manager

- Allows users to schedule tasks to run at different times of day/month/year.
 - Allows email notification upon completion of task.
 - Allows results to be directed to output file.

Task Name	Task Type	Namespace	Description
Switch Journal	System	%SYS	Switches the journal file at midnight every day
Purge Journal	System	%SYS	Purges old journal files every night at 12:30 am
Purge Tasks	System	%SYS	Purges the task history global every night at 1:00 am
Integrity Check	System	%SYS	Integrity check for databases at 2:00 am every Monday
Security Scan	System	%SYS	Scans the security database at midnight every day
Diagnostic Report	System	%SYS	Send system diagnostic reports to WRC On Demand, and/or on a schedule
Purge Audit Database	System	%SYS	Purges old Audit information after Switch Journal is run



Email Settings

- Configure email settings used for notifications from Task Manager.
 - Success and failure messages.
 - System Administration > Configuration > Additional Settings > Task Manager Email.

SMTP Server			
Port		SSL Config	☒ Use STARTTLS ☐
SMTP Auth User		Password	
Sender			
Reply To			
Success Subject	Scheduler process #(NAME)# - #(ID)# #(SUCCESS)#		
Success Message	Scheduler process #(NAME)# - #(ID)# Scheduled for #(SCHEDULED)# on #(SERVER)#: #(INSTANCE)#, #(SUCCESS)# Executing Code #(TASKCLASS)# at #(LASTFINISHED)# #(ERROR)#		
Failure Subject	Scheduler process #(NAME)# - #(ID)# #(SUCCESS)#		
Failure Message	Scheduler process #(NAME)# - #(ID)# Scheduled for #(SCHEDULED)# on #(SERVER)#: #(INSTANCE)#, #(SUCCESS)# Executing Code #(TASKCLASS)# at #(LASTFINISHED)# #(ERROR)#		



How To: Schedule Tasks

- System Operation > Task Manager > New Task.
- %SYS>do ^TASKMGR.
- %SYS.Task class.

The screenshot shows a configuration form for scheduling a task. The fields and their values are as follows:

Field	Value
Task name: *	
Description:	
Namespace to run task in:	%SYS
Task type: *	InventoryScan
Task priority:	Priority Normal
Run task as this user:	SuperUser
Open output file when task is running?	No
Output file:	
Suspend task on error?	No
Reschedule task after system restart?	No

Annotations on the form:

- A red oval highlights the "Run task as this user:" field, which is set to "SuperUser".
- A red arrow points from the "Reschedule task after system restart?" field (set to "No") to a text box on the right.

If task scheduled to run while system is down, should system reschedule it on restart?

- Yes - reschedule it to run at its next regular time after restart.
- No - run immediately on restart, and then return to regular schedule.



Task Types

- Pre-defined tasks exist for %SYS.
- Can create own tasks through class code.
 - Refer to %SYS.Task package for details on pre-defined tasks.
 - Refer to %SYS.Task.Definition class for details on creating own tasks.

The screenshot shows a task configuration form with the following fields and annotations:

- Task name:** * [text input field]
- Description:** [text input field]
- Namespace to run task in:** %SYS (dropdown menu, circled in red)
- Task type:** * InventoryScan (dropdown menu, circled in red)
- Task priority:** [dropdown menu]
- Run task as this user:** [dropdown menu]
- Open output file when task is running?** [checkbox]
- Output file:** [text input field]
- Suspend task on error?** [checkbox]
- Reschedule task after system restart?** [checkbox]

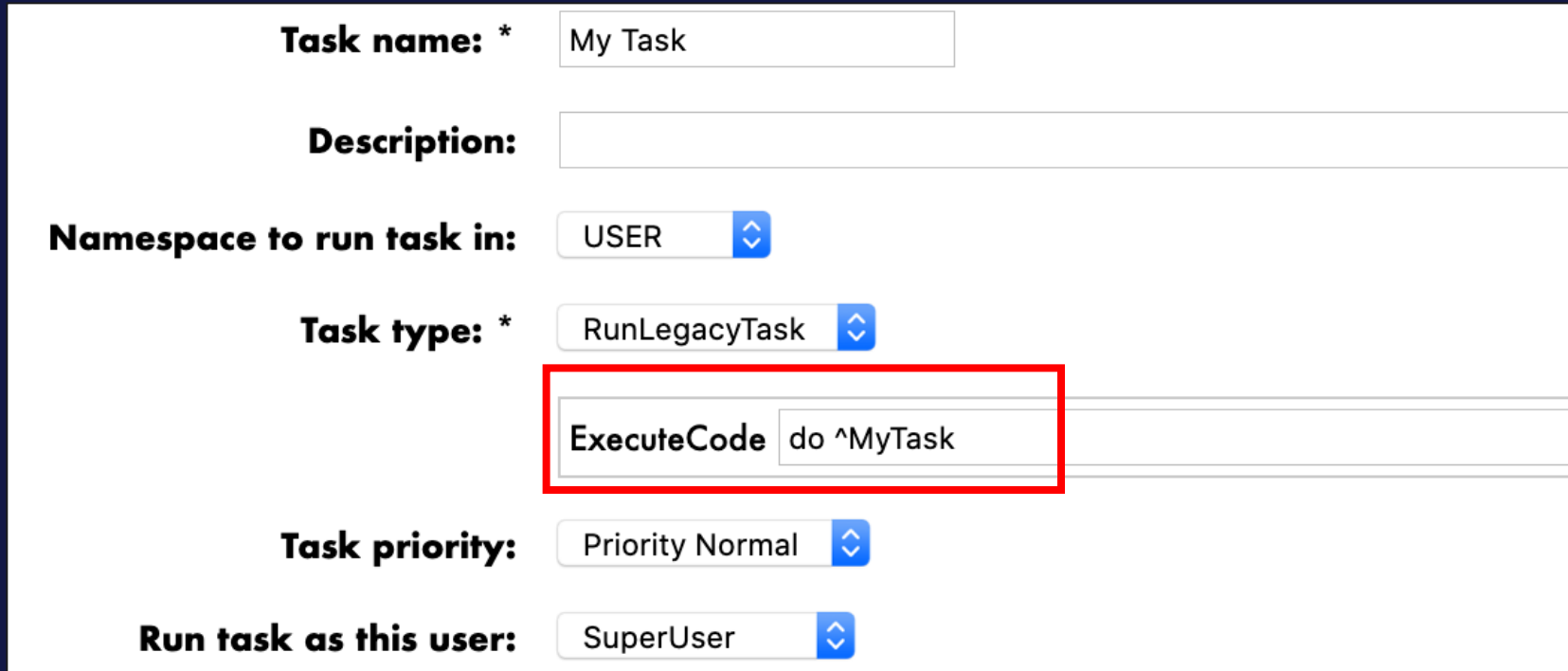
A red bracket on the right side of the form groups the following options in the Task type dropdown menu:

- %SYS.Task.FeatureTracker
- CheckLogging
- CleanSQLIndex
- Cube Manager Build
- Cube Manager Synch
- CumulIncrDBList
- DiagnosticReport
- FrozenPlanScan
- FullAllDatabases
- FullDBList
- IncrementalDBList
- IntegrityCheck
- ✓ InventoryScan
- Purge ZEN Reports
- PurgeAudit
- PurgeBackupLog
- PurgeErrorsAndLogs
- PurgeJournal



RunLegacyTask Task Type

- To enter code to execute instead of defined task, choose Task Type = RunLegacyTask.



The screenshot shows a task configuration form with the following fields:

- Task name: *** My Task
- Description:**
- Namespace to run task in:** USER
- Task type: *** RunLegacyTask
- ExecuteCode** do ^MyTask (This row is highlighted with a red box)
- Task priority:** Priority Normal
- Run task as this user:** SuperUser



How To: Manage Tasks

- System Operation > Task Manager.
- %SYS>do ^TASKMGR.
- %SYS.Task class.

New Task	i
On-demand Task	i
Upcoming Tasks	i
Task Schedule	i
Task History	i
Import Tasks	i



%ZSTART and %ZSTOP

- Each has sections which can run custom code independently.
 - SYSTEM - Start/stop of system.
 - LOGIN - User logs in or out.
 - JOB - Start/stop of background process.
 - Includes ODBC and cspServer processes.
 - CALLIN - Start/stop of process calling into system using CALLIN interface.



Controlling %ZSTART/%ZSTOP

- System Administration > Configuration > Additional Settings > Startup.

The diagram illustrates the mapping of configuration sections to specific settings. On the left, a grey box labeled '%ZSTART Sections' has red arrows pointing to 'CallinStart', 'JobStart', 'ProcessStart', and 'SystemStart'. On the right, a white box labeled '%ZSTOP Sections' has red arrows pointing to 'CallinHalt', 'JobHalt', 'ProcessHalt', and 'SystemHalt'.

Setting	Value	Edit	Help
ArbiterURL		Edit	Help
CallinHalt	true	Edit	Help
CallinStart	true	Edit	Help
CliSysName		Edit	Help
DBSizesAllowed	8192	Edit	Help
JobHalt	true	Edit	Help
JobServers	0	Edit	Help
JobStart	true	Edit	Help
LicenseID		Edit	Help
MaxConsoleLogSize	5	Edit	Help
ProcessHalt	true	Edit	Help
ProcessStart	true	Edit	Help
ShardClusterURL		Edit	Help
ShardMasterRegexp	-0\$	Edit	Help
ShardMirrorMember		Edit	Help
ShardRegexp	-[0-9]+\$	Edit	Help
ShardRole		Edit	Help
ShutdownTimeout	300	Edit	Help
SystemHalt	true	Edit	Help
SystemMode		Edit	Help
SystemStart	true	Edit	Help



Emergency Access Mode Considerations

- Task Manager does not start.
- %ZSTART and %ZSTOP do not run.



\$SYSTEM

- `$SYSTEM` provides access to utilities that:
 - Return information, such as:
 - # available license units.
 - # CPUs on system.
 - Or, perform operation, such as:
 - Upgrade license.
 - Change priority.
- `do $SYSTEM` when performing operation.
- `write $SYSTEM` when returning information.
- Some functionality only available using `$SYSTEM`.
- Well-documented in *Class Reference Information*.



\$SYSTEM.class.method()

- write \$SYSTEM.License.LUAvailable()
 - Class names and method names case sensitive.

The image shows a screenshot of a software interface. On the left is a tree view of the \$SYSTEM class hierarchy. A red box labeled '\$SYSTEM' points to the root of this tree. On the right is a table listing the methods of the \$SYSTEM class. A red box labeled 'Method' points to the 'LUAvailable' method in the table. A red box labeled 'Class' points to the 'License' class in the tree view. A red arrow points from the 'License' class in the tree view to the 'LUAvailable' method in the table.

\$SYSTEM

Method

Class

ActiveServerCount	ApplicationServerLogin	CKEY
CSPUsers	CheckAppLicenseUsed	CheckKeyForUpgrade
ConnectionCount	Decode	DecodeAuth
DumpKey	DumpKeys	DumpLocalAll
DumpLocalPID		DumpServers
GetAppLicenseFree	GetAppLicenseMin	GetAppLicenseUsed
GetAppNames	GetAppLicenseLimit	GetFeature
GetKeyStatus		GetUserLimit
InstanceConnection	InstanceConnectionsMax	IsAppLicensed
IsValidKey	KeyAuthorizationKey	KeyCPUsEnforced
KeyCoresEnforced	KeyCoresLicensed	KeyCustomerName
KeyExpirationDate	KeyFeatures	KeyFileType
KeyLicenseType	KeyLicenseUnits	KeyMachineID
KeyPlatform	KeyProduct	KeyServer
LUAvailable	LUConsumed	LUMaxConsumed
LUMinAvailable	LicenseCount	LoadAppLicenseFile
MaxBufferPool	MaxCSPGrace	MaxCSPUsers
MaxDB	MaxECPServers	NetworkEnabled
ProcConsumed	ProcMaxConsumed	ProcMinAvailable
PublicWebAppUserCount	PublicWebAppUsersMax	ReadDurationCounts
ReturnAllCSPApps	ReturnApplicationLicense	SetConnectionLimit
ShowCounts	ShowServer	ShowSummary
Upgrade	UserNameLicensing	



\$SYSTEM Examples

Class

Method

```
%SYS>w $SYSTEM.Util.NumberOfCPUs ()  
2  
%SYS>w $SYSTEM.SYS.ProcessID ()  
5996  
%SYS>w $SYSTEM.License.LUAvailable ()  
9  
%SYS>
```



Calling Utilities from OS Command Line or Script

- From command line:
 - `irissession InstanceName -U Namespace "Code"`
- *Code* **examples:**
 - `^ROUTINE`
 - `TAG^ROUTINE`
 - `##class (Package.Class).Method()`
 - `$SYSTEM.Class.Method()`
- **Must authenticate with system.**



^OPER

```
%SYS>do ^OPER
```

Operator's Utilities

1. Dismount databases (^DISMOUNT)
2. Mount databases (^MOUNT)
3. Lock Table Management (^LOCKTAB)
4. Free Count (ALL^%FREECNT)
5. Journal Utilities (^JOURNAL)
6. Backup Utilities (^BACKUP)
7. Examine Job (^JOBEXAM)
8. Terminate Job (^RESJOB)
9. Switch Set/Clear (^SWSET)
10. Free Up Terminal (^TTYFREE)
11. Broadcast a Message (^BROADCAST)
12. Display System Error Log (^SYSLOG)
13. Exit This Utility

Option?



Suggested Readings

- *System Administration Guide > Configuring Task Manager Email Settings.*
- *ObjectScript Reference > ObjectScript Special Variables > \$SYSTEM.*



Summary

- What are the key points for this module?

